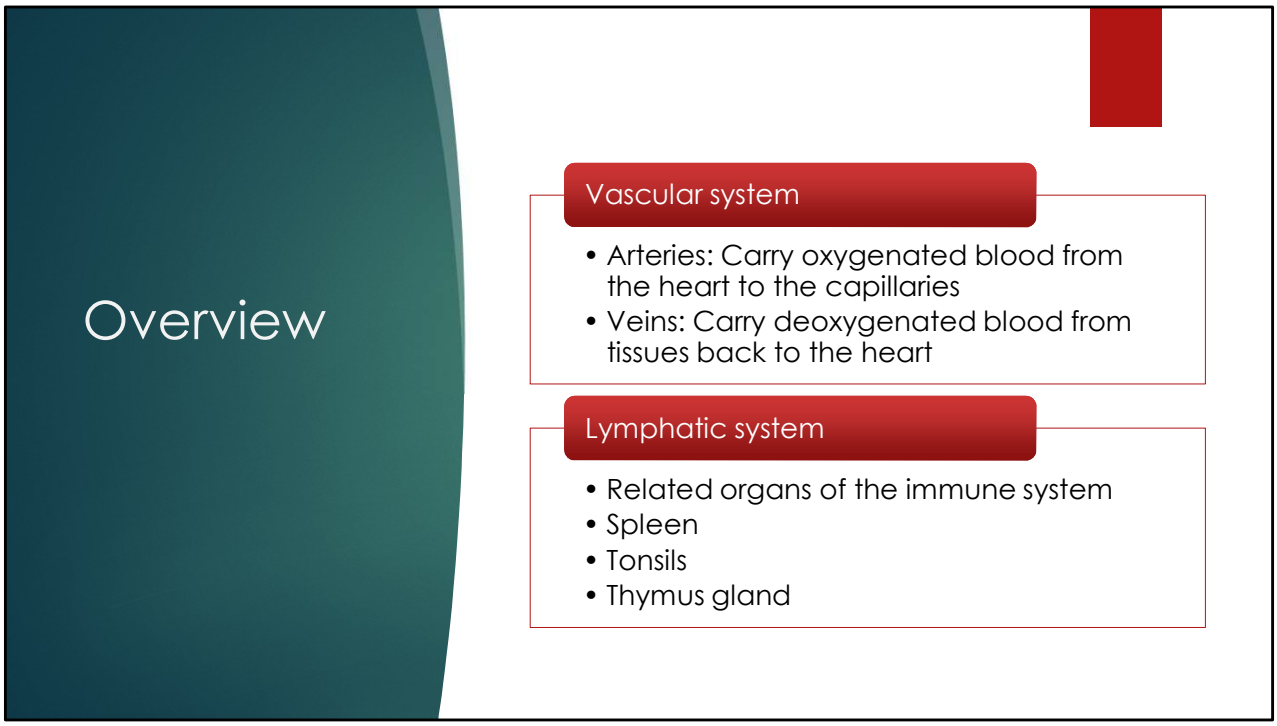
Assessing peripheral vascular system

CHAPTER 22 WEBER

NURS3120: HEALTH ASSESSMENT AND PROMOTION

Objectives

- Describe the structure and function of the blood vessels, including capillaries and lymphatic circulation
- Discuss risk factors associated with peripheral vascular disease and ways to reduce one's risks
- Differentiate between normal and abnormal findings of the peripheral vascular system
- Analyze the data from the interview and physical assessment of the peripheral vascular system to formulate valid clinical judgments

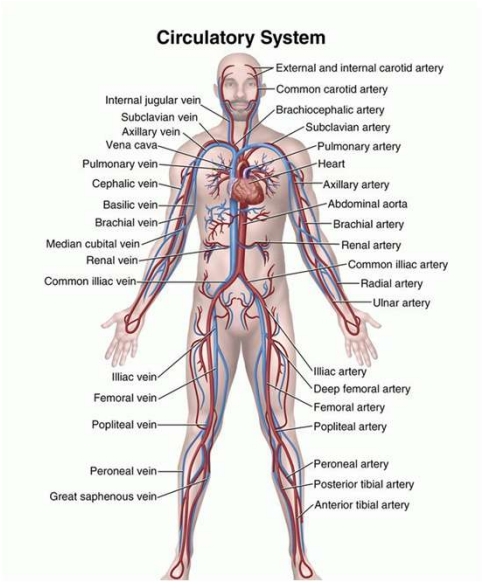


The arterial network is a high-pressure system. Blood is propelled under pressure from the left ventricle of the heart. Because of this high pressure, arterial walls must be thick and strong; the arterial walls also contain elastic fibers so that they can stretch. Each heartbeat forces blood through the arterial vessels under high pressure, creating a surge. This surge of blood is the arterial pulse.

The veins contain nearly 70% of the body's blood volume. Because blood in the veins is carried under much lower pressure than in the arteries, the vein walls are much thinner. In addition, veins are larger in diameter than arteries and can expand if blood volume increases. This helps to reduce the workload on the heart. There is no force that propels forward blood flow is the venous system. This fact is of special concern in the veins of the leg. Blood from the legs and lower trunk must flow upward with no help from the pumping action of the heart. It flows from actions of valves, skeletal muscle contraction, and the pressure gradient created with breathing.

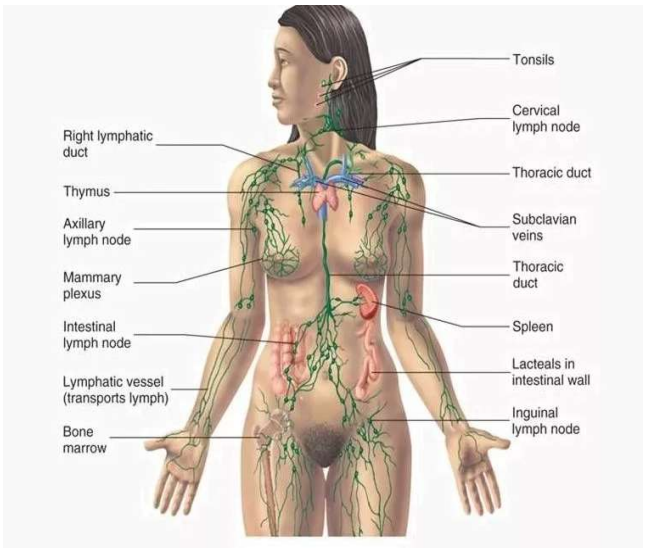
Spleen: left upper quad of the abdomen.
Spleen- destroys old RBC's, produce antibodies, store RBCs, filter microorganisms from the blood.

Tonsils: Entrance to the respiratory tract.
Tonsils- respond to local inflammation
Thymus gland: superior mediastinum behind the sternum in front of the aorta
Thymus- develops T lymphocytes in the immune system of children, atrophies after puberty.



Peripheral vascular system

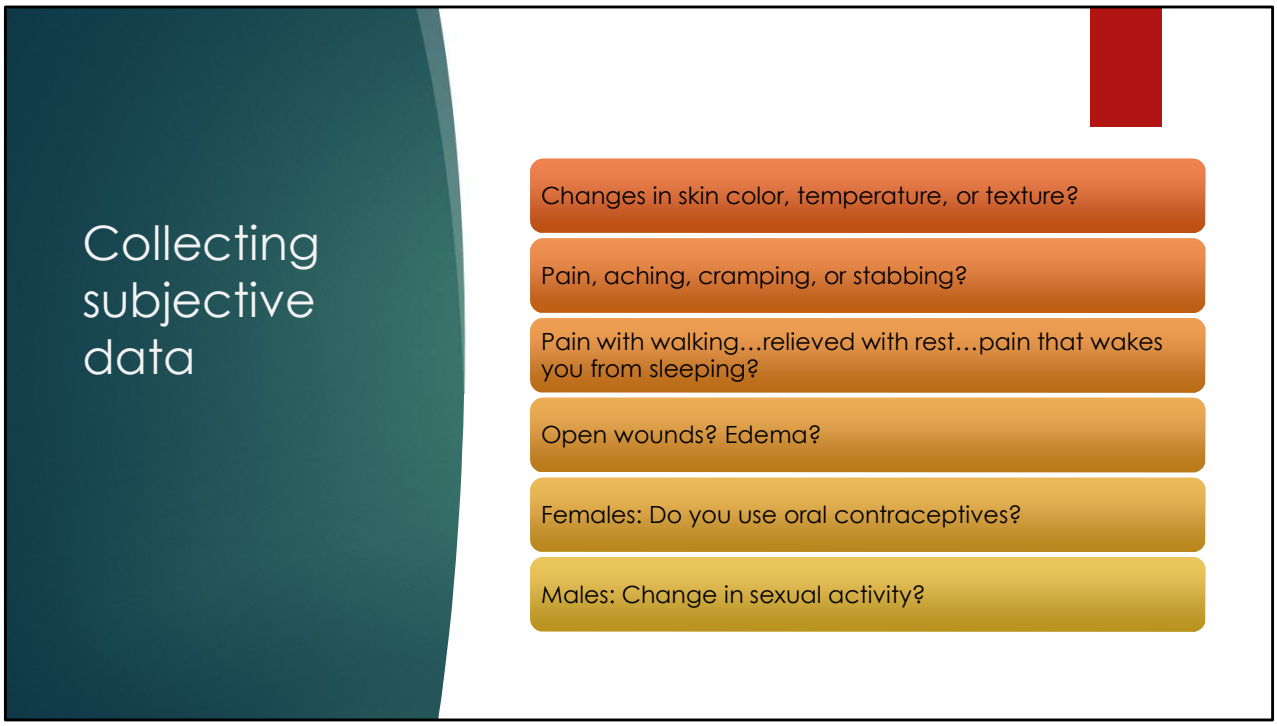
This is a complete picture of your circulatory system



Lymphatic system

The lymphatic system an integral and complementary component of the circulatory system, is a complex vascular system composed of lymphatic capillaries, lymphatic vessels and lymph nodes. Its PRIMARY function is to drain excess fluid and plasma proteins from bodily tissues and return them to the venous system. During circulation, more fluid leaves the capillaries than the veins can absorb. Draining excess fluid action prevents edema, which is a buildup of fluid in the interstitial spaces.

So with this information; does it make sense why a client with a mastectomy can not have blood pressures taken on their effected arm?



Color, temperature or texture changes? Cold, pale, clammy skin on the extremities and think, shiny skin can be associated with arterial insufficiency. Warm skin, edema, and brown pigmentation around the ankles are associated with venous insufficiency.
Pain or cramping? PAD blockage develops over a lifetime and often symptoms do not occur or appear until 60% blockage.
Intermittent claudication is usually the first symptom and is characterized by weakness, cramping, aching, fatigue, or frank pain with activity; located in the calves, thighs, or buttocks but rarely in the feet. These symptoms are relieved with rest. Other symptoms: burning or aching pain, cool skin, redness or other color changes of the skin, increased occurrence in infection, toe and foot sores that do not heal.
So what do you think is the pathophysiology of intermittent claudication? It is blood supply that is blocked or partial blocked and not able to flow down the leg sufficiently. So the supply of oxygen to the muscles/tissues is not sufficient.

Edema- swelling in your legs or feet is known as peripheral edema and results from an obstruction of the lymphatic flow or from venous insufficiency from such conditions as incompetent valves or decreased osmotic pressure in the capillaries.
Open sores- PVD is often associated with delayed wound healing. Ulcers associated

with arterial disease are usually painful and are often located on the toes, foot, or lateral ankle. Venous ulcers are usually painless and occur on the lower leg or medical ankle. We will go more in depth later in the powerpoint.
Female: Do you take contraceptives? What if they smoke and take contraceptives? DVT risk.
Male: Decrease in your usual sexual activity? Central arterial or venous disease may be manifested early as erectile dysfunction ED. ED may occur with decrease blood flow or an occlusion of the blood vessels in one type of PAD known as aortoiliac occlusion. Men be more reluctant to report or discuss difficulties achieving or maintaining an erection.

Health promotion and disease prevention: Peripheral artery disease (PAD)

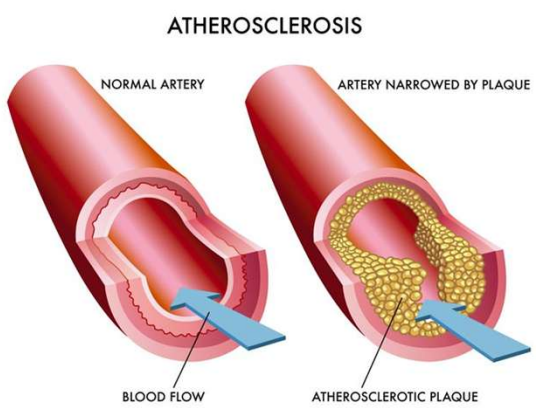
- AKA: Peripheral vascular disease
- "Hardening of the arteries"
- Blood vessels become thick & stiff
  - Restricts blood flow
- Peripheral blood vessels are more rigid with age



Arteriosclerosis

PAD

- Develops gradually
- Build up of fatty deposits in the artery walls
- Build up causes arteries to narrow, thus reducing blood flow
- Common
- Asymptomatic in many individuals
- Increases with smokers and diabetics



In PAD, the arteries slowly become narrowed or blocked when plaque gradually forms inside the artery walls, which causes blood flow to organs and tissues to be reduced or stopped, resulting in damage. If left untreated, the individual is at risk for heart attack, stroke, transient ischemic attack (TIA), renal artery disease or stenosis, and amputation.

20-50% asymptomatic.
Atypical leg pain, painful walking, poor wound healing.
Worse with smoking and/or diabetes.
Other risk factors: Coronary heart disease, obesity, HTN, high cholesterol, family history, African American, high levels of homocysteine. (Homocysteine is an amino acid in the blood produced during protein metabolism, usually broken down by vitamins B6, B12, and folate. Elevated levels indicate potential B-vitamin deficiencies, kidney disease, or genetic factors, and are **associated with increased risks of cardiovascular disease, stroke, blood clots, and dementia.)**

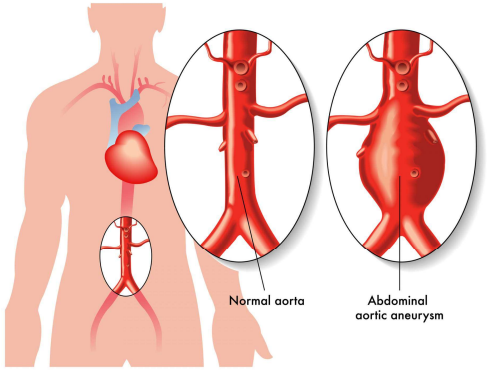
PAD



- Client education
  - Stop smoking
  - Control blood glucose
  - Regular exercise
  - Maintain normal cholesterol levels
  - Healthy diet
  - Maintain healthy weight
  - Higher risk for DVT, stroke, and poor wound healing
    - Teach signs and symptoms

Abdominal aortic aneurysm: AAA

- An enlargement of the aorta, the main blood vessel that delivers blood to the body at the level of the abdomen
- Very common
- Often grows slowly without symptoms
- If the aneurysm is growing larger, pulsations at the abdomen may be palpable or visible
- A ruptured aneurysm is life-threatening



(Covered in chapter 23 in the text)
Small less than 5.5 centimeters or slow growing
Large: greater than 5.5 centimeters or fast-growing. Large AAA are more likely to rupture than small or slow growing
Causes of AAA? Smoking, high blood pressure, vasculitis
Risk factors: male, obese or overweight, over age 60, family history of heart conditions, and diseases, high blood pressure, high cholesterol, sedentary lifestyle, smoker.

So if you have a patient that has a known AAA- sudden pain in your abdomen and back, clammy, sweaty skin, increased heart rate, loss of consciousness might be a ruptured AAA.

Inspection and palpation of the arms



Inspect:

- Color, size, and wounds
- Nails

Palpate:

- Temperature
- Capillary refill
- Brachial, radial, and ulnar pulses
- Edema
- Perform the Allen test

Pallor, cyanosis, redness, swelling, pain, numbness, tingling burning, throbbing, and coldness. Lymphedema will cause unilateral edema.

Raynaud disorder: A vascular disorder caused by vasoconstriction or vasospasms of the fingers or toes, characterized by rapid changes of color (pallor, cyanosis, and redness) swelling, pain, numbness, tingling, burning, throbbing, and coldness. The disorder commonly occurs bilaterally; symptoms last minutes to hours. Raynaud's affects up to 1 in 20 people in the United States, more often in people assigned female at birth and in people under age 30.

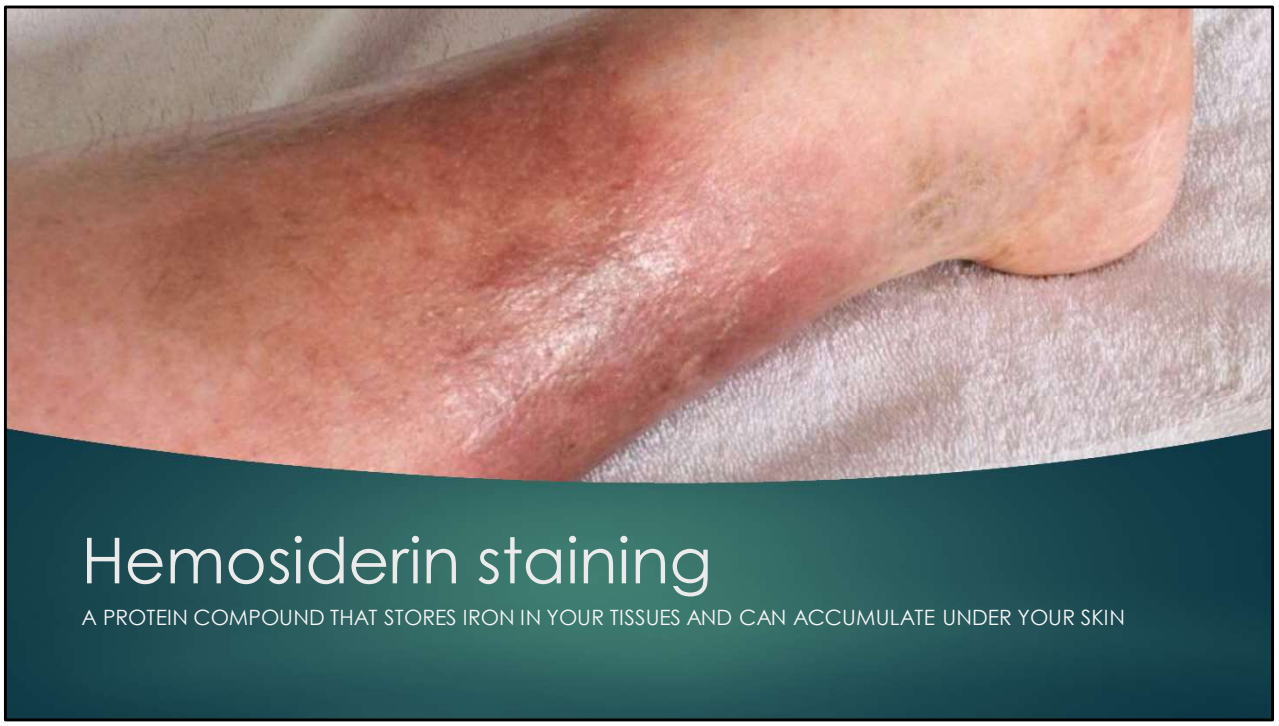
Cap refill: Capillary refill time exceeding 2 seconds may indicate vasoconstriction, decreased cardiac output, shock, arterial occlusion, or hypothermia.

The Allen Test is to evaluate patency of the radial or ulnar arteries. Have the client rest the hand palm side up on the examination table and make a fist. Then use your thumbs to occlude the radial and ulnar arteries. Continue pressure to keep both arteries occluded and have the client relax the fist. Note that the palm remains pale. Release the pressure on the ulnar artery and each for color to return to the hand. Repeat and release the radial artery. Abnormal would be if pallor persists and

does not return to the palms within 3-5 seconds.



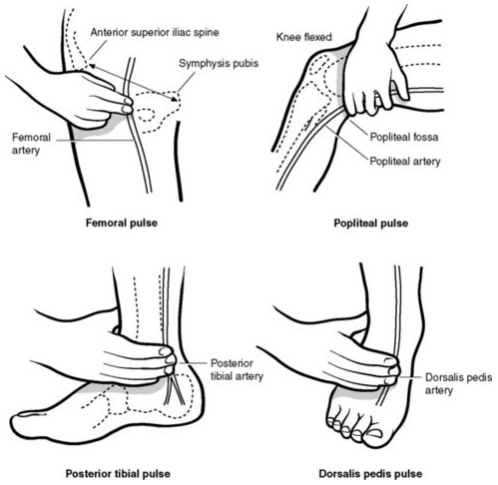
Abnormal findings-pallor, dark-colored toes and blisters are seen with arterial insufficiency and gangrene.
Edema: can be caused by many things- heart failure, constriction of the blood vessels, DVT
Loss of hair on the legs suggests arterial insufficiency. Often thin, shiny skin is noted
Ulcers: wounds, lesions
Gangrene
Thick-ridged nails
Symmetry of the legs
Brown discoloration- chronic venous stasis caused by hemosiderin deposits from RBCs degeneration.



Hemosiderin a protein compound that stores iron in your tissues and can accumulate under your skin. When red blood cells break down, the hemoglobin releases iron. The trapped iron is then stored as hemosiderin in tissues beneath your skin, causing visible hemosiderin staining.

Palpate the arteries of the legs

- Femoral
- Popliteal
- Posterior tibial
- Dorsalis pedis



Femoral - Just below the inguinal ligament and halfway between the pubis and anterior superior iliac spines
Popliteal – the popliteal fossa behind the knee
Posterior tibial – curve fingers around the medial malleolus and the Achilles tendon
Dorsalis Pedis – light touch lateral and parallel with the extensor tendon of the big toe.

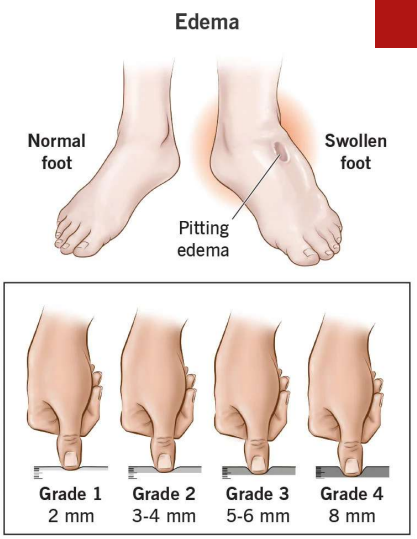
The doppler ultrasound device



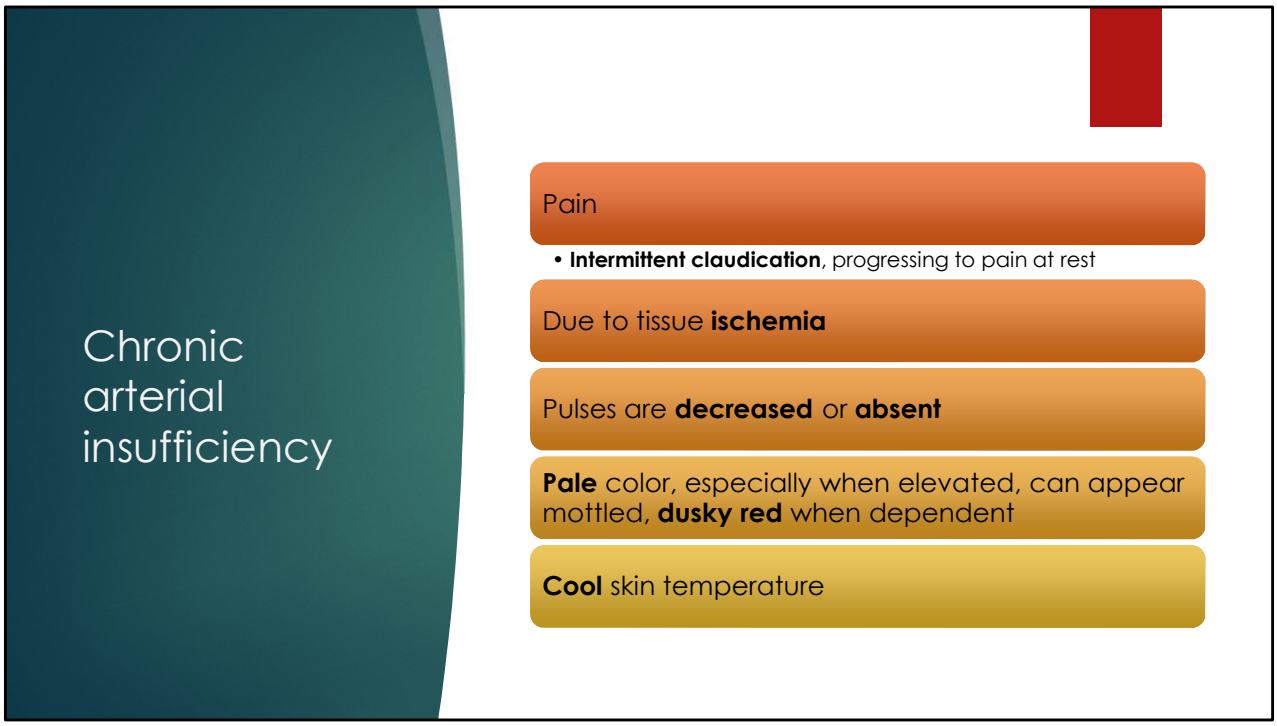
- Transmits and receives ultrasound waves to evaluate blood flow
- Used to:
  - Assess unpalpable pulses in the extremities
  - Determine the patency of arterial bypass grafts
  - Assess tissue perfusion in an extremity

Palpate edema

- Rated on a scale 0-4
- 0: No edema
- 1+: Mild
- 2+: Moderate
- 3+: Deep
- 4+: Very deep



0: No edema
1+ mild pitting, slight indention, no perceptible swelling.
2+ Moderate pitting, indention subsides 10-15 seconds
3+ Deep pitting, indention remains for >1 minute, legs look swollen
4+ very deep pitting, indention lasts >2 minutes, leg is grossly swollen and distorted



Pain: Intermittent claudication to sharp, unrelenting, constant.

Chronic arterial insufficiency

- Edema may be **absent** or **mild**
- Skin appears **shiny**, **thin**, **dry**, **loss of hair** on the lower leg/foot
- Nails are **thick** and **ridged**
- Ulcerations generally on **toes**, **heels**, or **points of trauma**
- Ulcers are **very painful** and deep
- Shape is **circular** with well-defined edges
- Ulcer base is pale black to dry and **gangrene**



Arterial (Ischemic) ulcer- due to decreased blood flow to the area, tend to be on the toes, metatarsal heads, heels, and lateral ankle. Pale, ischemic base, well-defined edges

Chronic venous insufficiency

- Pain is **aching/cramping**
- Due to venous stasis and hypertension
- Pulses tend to be **normal**, but **difficult to palpate** due to profound edema
- Skin temperature is **normal**



Account for 80% of lower leg ulcers
Bleeding, uneven edges
Hemosiderin changes
There is a great chart on page 490 (Box 22-1) that lists the difference between arterial insufficiency and venous insufficiency please review that

Chronic venous insufficiency

- Edema is **pronounced**
- Skin
  - **Thick** and **rough**
  - **Hemosiderin staining**
  - Stasis **dermatitis**
- Ulcerations tend to occur at the **sides of the ankle**, especially medially
- Ulcer depth is **superficial**
  - Base is **beefy red** to **yellow**

Neuropathic ulcers

- Develops in pressure point areas with diminished sensation
  - Often in diabetics with diabetic neuropathy
- There is no pain associated with this so the ulcer may go unnoticed for some time
- Increased risk for infection



Related to diabetes, dysfunction of the arterial wall



Abnormal arterial findings

Gangrene which is death of the tissue caused by infection or lack of blood flow.

Reynaud's on the right

Abnormal venous findings

- Lymphedema
- Blockage of the lymphatic system
- Classified into stages
  - 0: Subclinical stage (Can be months or years before progressing)
  - 1: Mild
  - 2: Moderate
  - 3: Severe



Is blockage of the lymphatic system, part of the immune and circulatory systems. Lymphedema is classified into stages: 0=No obvious signs or symptoms. Impaired lymph drainage is subclinical. Edema is not evident.
Stage 1: Swelling is present. Affected area pits with pressure. Elevation relieves swelling. Skin texture is smooth (Spontaneously reversible)
Stage 2: Tissue is firmer. Skin may look tight, shiny, and tissue may have a spongy feel. Pitting may or may not be present as tissue fibrosis (hardening) begins to develop. Elevation does not completely alleviate the swelling. Hair loss or nail changes may be experienced in affected extremity. Assistance will be needed to reduce edema.
Stage 3: Irreversible: The extremity affected has progressed to the lymphostatic elephantiasis stage, at which the limb is very large. Affected area is non-pitting, often with permanent eczema. Skin is firm and thick with hard fibrotic underlying tissue having an unresponsive feel. Skin folds develop. At increased risk for recurrent cellulitis, (skin infection). Affected area may ooze fluid. Elevation will not alleviate symptoms.



Abnormal venous findings

Varicose veins & spider veins

Varicose Veins are hereditary but may also develop from increased venous pressure and venous pooling. Standing in one place for a long period of time increases the risk for varicosities. So please wear ted hose. The older female client may be more self aware and self conscious of these veins.

Nursing School exam questions be like...

A 47-yo man presents with a diffuse maculopapular rash. He has a history of hypertension, for which he takes these 15 medications. Assuming there are seven monkeys on the roof and two in the garden, what is the second enzyme required to make the color purple?